Validation Report: Traceability of the FCA Formal Context to EO Discourse

Concept: Thing | Source: Dietz & Mulder, Enterprise Ontology (2nd ed., Springer 2024)

1. Summary Table

Check 1 – Items

Item Label	Reference Key	Outcome
54-01 a thought	54-01	PASS
56-05 a cloud	56-05	PASS
56-12 a body	56-12	PASS
56-19 a human	56-19	PASS
56-19 a dog	56-19	PASS
56-19 a membership	56-19	PASS
56-19 a rental	56-19	PASS
56-20 a number	56-20	PASS
56-20 a boolean value	56-20	PASS
56-27 humans	56-27	PASS
56-27 rentals	56-27	PASS
56-33 <the human Charlie Chaplin>	56-33	PASS
56-33 (the human Charlie Chaplin)	56-33	PASS
57-02 [HUMAN]	57-02	PASS
57-03 (the class HUMAN)	57-03	PASS
58-06 (the number 387)	58-06	PASS
59-02 Charlie Chaplin	59-02	PASS
59-04 day 2411108	59-04	PASS
59-08 a car	59-08	PASS
59-08 a house	59-08	PASS
59-08 an insurance policy	59-08	PASS
59-08 an employment contract	59-08	PASS
60-10 [some thing]	60-10	PASS
60-21 HUMAN	60-21	PASS
60-25 (HUMAN)	60-25	PASS
62-21 a student	62-21	PASS
62-24 cars	62-24	PASS
62-24 boats	62-24	PASS
62-24 aircrafts	62-24	PASS
62-24 vehicles	62-24	PASS
65-02 this chair	65-02	PASS
66-05 a ship	66-05	PASS
66-14 a butterfly	66-14	PASS
66-14 a caterpillar	66-14	PASS
66-25 a human body	66-25	PASS

Check 2 – Characteristics

Characteristic Label	Reference Key	Outcome
54-01 in the mind	54-01	PASS
54-02 outside the mind	54-02	PASS
55-12 collection of Things	55-12	PASS
55-12 untouched by mental abstractions	55-12	PASS
56-01 distinguishable	56-01	PASS
56-01 collection of Concrete things	56-01	PASS
56-02 Object with assigned form	56-02	PASS
56-14 with Object	56-14	PASS
56-19 with Concrete object	56-19	PASS
56-20 with Abstract object	56-20	PASS
56-21 can be observed	56-21	PASS
56-21 can be represented by Conceptual thing	56-21	PASS
56-21 represents a Concrete thing	56-21	PASS
56-24 member of a Class	56-24	PASS
56-28 with a Conceptual object	56-28	PASS
56-29 can conform to a Type	56-29	PASS
56-29 member of a conceived world	56-29	PASS
56-30 instance of a Type	56-30	PASS
56-32 collection of strings	56-32	PASS
56-33 can be represented by a Conceptual thing	56-33	PASS
57-09 collection of numeric codes	57-09	PASS
58-14 variable of Conceptual fact	58-14	PASS
59-01 variable of Unary fact	59-01	PASS
59-02 variable of Concrete unary fact	59-02	PASS
59-04 of time scale	59-04	PASS
59-08 in reality	59-08	PASS
59-09 tangible	59-09	PASS
59-09 intangible	59-09	PASS
60-05 can be an instance of a Type	60-05	PASS
60-06 with a Form	60-06	PASS
60-08 may conform to a Type	60-08	PASS
60-08 member of a World	60-08	PASS
60-21 variable of an Unary fact	60-21	PASS
60-22 extension of a Type	60-22	PASS
60-23 with type intension	60-23	PASS
60-25 represents a Concrete class	60-25	PASS
60-25 can be represented by a Conceptual class	60-25	PASS
61-04 variable of a Fact	61-04	PASS
61-06 comes into existence	61-06	PASS
61-06 part of an event	61-06	PASS
61-09 with Form	61-09	PASS
61-12 goes out of existence	61-12	PASS
65-01 can be conceptualised	65-01	PASS
65-02 can have a role in a relationship	65-02	PASS
66-05 a collection of components	66-05	PASS

69-06 with properties	69-06	PASS
54-01 Thing	54-01	PASS
54-02 Concrete thing	54-02	PASS
55-12 World	55-12	PASS
56-20 Abstract thing	56-20	PASS
56-21 Conceptual thing	56-21	PASS
56-24 Class	56-24	PASS
56-29 Tangible concrete thing	56-29	PASS
56-29 Intangible concrete thing	56-29	PASS
56-29 Concieved world	56-29	PASS
56-32 Name class	56-32	PASS
57-02 Concrete class	57-02	PASS
60-21 Conceptual class	60-21	PASS
66-05 Composite thing	66-05	PASS

Check 3 – Relationships (Explicit References)

Item — Characteristic	Cell Reference	Outcome
56-19 a human — 56-21 can be observed	56-29	PASS
56-19 a human — 59-09 tangible	56-29	PASS
56-19 a human — 54-01 Thing	59-08	PASS
56-19 a dog — 56-21 can be observed	56-29	PASS
56-19 a dog — 59-09 tangible	56-29	PASS
56-19 a membership — 59-09 intangible	56-29	PASS
56-19 a rental — 59-09 intangible	56-29	PASS
56-19 a rental — 54-01 Thing	59-08	PASS
56-33 (the human Charlie Chaplin) — 56-30 instance of a Type	58-14	PASS
59-02 Charlie Chaplin — 54-01 Thing	65-02	PASS
59-08 a car — 66-05 a collection of components	66-26	PASS
59-08 a car — 66-05 Composite thing	66-26	PASS
59-08 a house — 56-29 Tangible concrete thing	59-09	PASS
60-21 HUMAN — 56-24 Class	60-23	PASS
66-05 a ship — 56-14 with Object	66-13	PASS
66-05 a ship — 56-19 with Concrete object	66-19	PASS
66-05 a ship — 60-05 can be an instance of a Type	66-10	PASS
66-05 a ship — 60-08 may conform to a Type	66-14	PASS
66-05 a ship — 54-02 Concrete thing	66-19	PASS
66-14 a butterfly — 56-30 instance of a Type	61-06	PASS
66-14 a caterpillar — 56-30 instance of a Type	61-12	PASS

Check 4 – Completeness Findings

Type	Entry	Discourse Reference
MISSING RELATIONSHIP	(the class HUMAN) — Conceptual thing	57-03
MISSING ITEM	business entity (or: business thing)	68-09

2. Check 1 Detail – Items with References

Objective: Verify that every item in G with an explicit discourse reference can be traced to the concept expressed or implied by the cited discourse sentence.

ITEM: 54-01 a thought

Reference: 54-01

Discourse sentence: "Thoughts are things in the mind."

Concept traceable: YES

Explanation: Sentence 54-01 explicitly introduces 'a thought' as an example of a Thing in the mind, matching the item's label.

Outcome: **PASS**

ITEM: 56-05 a cloud

Reference: 56-05

Discourse sentence: "The cloud is a Thing, of which the main Features are the recognisable water drops or crystals."

Concept traceable: YES

Explanation: Sentence 56-05 explicitly names 'The cloud' as a canonical example of a Thing, establishing the cloud as an item.

Outcome: **PASS**

ITEM: 56-12 a body

Reference: 56-12

Discourse sentence: "Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body."

Concept traceable: YES

Explanation: Sentence 56-12 uses 'a body' as a concrete example of a Thing whose Features change over time, directly naming the item.

Outcome: **PASS**

ITEM: 56-19 a human

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-19 explicitly lists 'humans' as one of the canonical Concrete things with Concrete objects, directly grounding the item.

Outcome: **PASS**

ITEM: 56-19 a dog

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-19 lists 'dogs' among the canonical Concrete things with Concrete objects.

Outcome: **PASS**

ITEM: 56-19 a membership

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-19 lists 'memberships' as a canonical Concrete thing with a Concrete object.

Outcome: **PASS**

ITEM: 56-19 a rental

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-19 lists 'rentals' as a canonical Concrete thing with a Concrete object.

Outcome: **PASS**

ITEM: 56-20 a number

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: YES

Explanation: Sentence 56-20 explicitly lists 'numbers' as Abstract things with Abstract objects.

Outcome: **PASS**

ITEM: 56-20 a boolean value

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: YES

Explanation: Sentence 56-20 explicitly lists 'boolean values' as Abstract things with Abstract objects.

Outcome: **PASS**

ITEM: 56-27 humans

Reference: 56-27

Discourse sentence: "As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals."

Concept traceable: YES

Explanation: Sentence 56-27 explicitly names 'humans' as a canonical example of a Class (Things belonging to a Class of the same type).

Outcome: **PASS**

ITEM: 56-27 rentals

Reference: 56-27

Discourse sentence: "As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals."

Concept traceable: YES

Explanation: Sentence 56-27 explicitly names 'rentals' as a canonical example of a Class.

Outcome: **PASS**

ITEM: 56-33 <the human Charlie Chaplin>

Reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Concept traceable: YES

Explanation: Sentence 56-33 explicitly introduces the notation <the human 'Charlie Chaplin'> as the concrete thing item.

Outcome: **PASS**

ITEM: 56-33 (the human Charlie Chaplin)

Reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Concept traceable: YES

Explanation: Sentence 56-33 explicitly introduces (the human 'Charlie Chaplin') as the corresponding conceptual thing item.

Outcome: **PASS**

ITEM: 57-02 [HUMAN]

Reference: 57-02

Discourse sentence: "The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans."

Concept traceable: YES

Explanation: Sentence 57-02 explicitly introduces the Concrete class [HUMAN] as the set of Concrete things that are humans.

Outcome: **PASS**

ITEM: 57-03 (the class HUMAN)

Reference: 57-03

Discourse sentence: "It is represented in our mind by the thought (the class HUMAN)."

Concept traceable: YES

Explanation: Sentence 57-03 states that (the class HUMAN) is the thought in the mind representing [HUMAN], introducing it as an item.

Outcome: **PASS**

ITEM: 58-06 (the number 387)

Reference: 58-06

Discourse sentence: "The name "the number '387'" signifies the abstract thing (the number '387')."

Concept traceable: YES

Explanation: Sentence 58-06 explicitly introduces the abstract thing (the number '387') by name.

Outcome: **PASS**

ITEM: 59-02 Charlie Chaplin

Reference: 59-02

Discourse sentence: "As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin"."

Concept traceable: YES

Explanation: Sentence 59-02 explicitly introduces 'Charlie Chaplin' as the canonical example of an entity (Concrete unary fact variable).

Outcome: **PASS**

ITEM: 59-04 day 2411108

Reference: 59-04

Discourse sentence: "As an example, day 2411108 is a value on the time scale Julian Date."

Concept traceable: YES

Explanation: Sentence 59-04 explicitly introduces 'day 2411108' as an example of an Abstract thing Value on the Julian Date time scale.

Outcome: **PASS**

ITEM: 59-08 a car

Reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Concept traceable: YES

Explanation: Sentence 59-08 explicitly names 'the car we drive in' as a Concrete thing in reality; by concept primacy, 'a car' refers to the same concept.

Outcome: **PASS**

ITEM: 59-08 a house

Reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Concept traceable: YES

Explanation: Sentence 59-08 explicitly names 'the house we live in' as a Concrete thing in reality; by concept primacy, 'a house' refers to the same concept.

Outcome: **PASS**

ITEM: 59-08 an insurance policy

Reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Concept traceable: YES

Explanation: Sentence 59-08 explicitly mentions 'the insurance policy for our car' as a Concrete thing in reality.

Outcome: **PASS**

ITEM: 59-08 an employment contract

Reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Concept traceable: YES

Explanation: Sentence 59-08 explicitly mentions 'the employment contract we have with our employer' as a Concrete thing in reality.

Outcome: **PASS**

ITEM: 60-10 [some thing]

Reference: 60-10

Discourse sentence: "Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin)."

Concept traceable: YES

Explanation: Sentence 60-10 explicitly introduces [some thing] as the notation for a Concrete thing studied in the semiotic mill example, directly grounding the item.

Outcome: **PASS**

ITEM: 60-21 HUMAN

Reference: 60-21

Discourse sentence: "Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN."

Concept traceable: YES

Explanation: Sentence 60-21 explicitly defines 'HUMAN' as the name of the Conceptual class of Things predicated by Facts of the Type human.

Outcome: **PASS**

ITEM: 60-25 (HUMAN)

Reference: 60-25

Discourse sentence: "Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN]."

Concept traceable: YES

Explanation: Sentence 60-25 explicitly introduces (HUMAN) as the conceptual class that represents the concrete class [HUMAN].

Outcome: **PASS**

ITEM: 62-21 a student

Reference: 62-21

Discourse sentence: "When this is not the case anymore, the student entity ceases to exist."

Concept traceable: YES

Explanation: Sentence 62-21 explicitly names 'the student entity' as a canonical example of a Concrete thing that ceases to exist (goes out of existence).

Outcome: **PASS**

ITEM: 62-24 cars

Reference: 62-24

Discourse sentence: "We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft."

Concept traceable: YES

Explanation: Sentence 62-24 explicitly names 'car' as one of the three types whose extensions are unified into the class vehicles.

Outcome: **PASS**

ITEM: 62-24 boats

Reference: 62-24

Discourse sentence: "We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft."

Concept traceable: YES

Explanation: Sentence 62-24 explicitly names 'boat' as one of the types whose extensions form the class vehicles.

Outcome: **PASS**

ITEM: 62-24 aircrafts

Reference: 62-24

Discourse sentence: "We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft."

Concept traceable: YES

Explanation: Sentence 62-24 explicitly names 'aircraft' as one of the types whose extensions form the class vehicles; 'aircrafts' is the pluralised form used in the formal context.

Outcome: **PASS**

ITEM: 62-24 vehicles

Reference: 62-24

Discourse sentence: "We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft."

Concept traceable: YES

Explanation: Sentence 62-24 explicitly introduces 'vehicle' as the type whose intension covers the union of cars, boats, and aircraft.

Outcome: **PASS**

ITEM: 65-02 this chair

Reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Concept traceable: YES

Explanation: Sentence 65-02 explicitly names 'this chair' as a Thing participating in the sit-on-ability relationship example.

Outcome: **PASS**

ITEM: 66-05 a ship

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: YES

Explanation: Sentence 66-05 introduces Theseus' ship as the canonical example of a Composite thing in the identity-change discussion; 'a ship' is the concept exemplified.

Outcome: **PASS**

ITEM: 66-14 a butterfly

Reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Concept traceable: YES

Explanation: Sentence 66-14 explicitly names 'a butterfly' as an example of a Thing that has kept its Object after transformation.

Outcome: **PASS**

ITEM: 66-14 a caterpillar

Reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Concept traceable: YES

Explanation: Sentence 66-14 explicitly names 'a caterpillar' as the prior form of the Thing that transforms into a butterfly, directly grounding the item.

Outcome: **PASS**

ITEM: 66-25 a human body

Reference: 66-25

Discourse sentence: "Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore."

Concept traceable: YES

Explanation: Sentence 66-25 explicitly names 'the human body' as a Composite thing whose identity is located in one of its components.

Outcome: **PASS**

3. Check 2 Detail – Characteristics with References

Objective: Verify that every characteristic in M with an explicit discourse reference can be traced to the concept expressed or implied by the cited discourse sentence.

CHARACTERISTIC: 54-01 in the mind

Reference: 54-01

Discourse sentence: "Thoughts are things in the mind."

Concept traceable: YES

Explanation: Sentence 54-01 states 'Thoughts are things in the mind', directly naming 'in the mind' as the defining characteristic of a thought.

Outcome: **PASS**

CHARACTERISTIC: 54-02 outside the mind

Reference: 54-02

Discourse sentence: "They represent concrete things (i.e. things outside the mind), called referents."

Concept traceable: YES

Explanation: Sentence 54-02 defines Concrete things as 'things outside the mind', directly naming the characteristic.

Outcome: **PASS**

CHARACTERISTIC: 55-12 collection of Things

Reference: 55-12

Discourse sentence: "Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions."

Concept traceable: YES

Explanation: Sentence 55-12 states a world is 'a collection of things', establishing this as a characteristic of World.

Outcome: **PASS**

CHARACTERISTIC: 55-12 untouched by mental abstractions

Reference: 55-12

Discourse sentence: "Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions."

Concept traceable: YES

Explanation: Sentence 55-12 states the world is 'untouched yet by mental abstractions', directly naming this characteristic.

Outcome: **PASS**

CHARACTERISTIC: 56-01 distinguishable

Reference: 56-01

Discourse sentence: "The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short."

Concept traceable: YES

Explanation: Sentence 56-01 states concrete things are 'distinguishable', directly naming this characteristic.

Outcome: **PASS**

CHARACTERISTIC: 56-01 collection of Concrete things

Reference: 56-01

Discourse sentence: "The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short."

Concept traceable: YES

Explanation: Sentence 56-01 states 'a world consists of distinguishable concrete things', establishing World as a collection of Concrete things.

Outcome: **PASS**

CHARACTERISTIC: 56-02 Object with assigned form

Reference: 56-02

Discourse sentence: "A Thing is said to be an Object with assigned Features."

Concept traceable: YES

Explanation: Sentence 56-02 explicitly states 'A Thing is said to be an Object with assigned Features'; by concept primacy, 'Features' and 'form' are the same concept.

Outcome: **PASS**

CHARACTERISTIC: 56-14 with Object

Reference: 56-14

Discourse sentence: "In this bare Object resides the identity of the thing."

Concept traceable: YES

Explanation: Sentence 56-14 states 'In this bare Object resides the identity of the thing', directly implying that a Thing has (is with) an Object.

Outcome: **PASS**

CHARACTERISTIC: 56-19 with Concrete object

Reference: 56-19

Discourse sentence: "Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-19 states 'Concrete objects are Objects of Concrete things', establishing that Concrete things have (are with) Concrete objects.

Outcome: **PASS**

CHARACTERISTIC: 56-20 with Abstract object

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: YES

Explanation: Sentence 56-20 states 'Abstract objects are Objects of Abstract things', establishing that Abstract things have Abstract objects.

Outcome: **PASS**

CHARACTERISTIC: 56-21 can be observed

Reference: 56-21

Discourse sentence: "When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing."

Concept traceable: YES

Explanation: Sentence 56-21 states 'When observing and distinguishing the Concrete things in a world', establishing that Concrete things can be observed.

Outcome: **PASS**

CHARACTERISTIC: 56-21 can be represented by Conceptual thing

Reference: 56-21

Discourse sentence: "When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing."

Concept traceable: YES

Explanation: Sentence 56-21 states 'the mind creates Conceptual things that represent the Concrete Things', establishing that Concrete things can be represented by Conceptual things.

Outcome: **PASS**

CHARACTERISTIC: 56-21 represents a Concrete thing

Reference: 56-21

Discourse sentence: "When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are

Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing."

Concept traceable: YES

Explanation: Sentence 56-21 states Conceptual things 'represent the Concrete Things', establishing this as a characteristic of Conceptual things.

Outcome: **PASS**

CHARACTERISTIC: 56-24 member of a Class

Reference: 56-24

Discourse sentence: "It is called classifying: we observe similarities and differences between things and accordingly put them into different classes."

Concept traceable: YES

Explanation: Sentence 56-24 states things are 'put into different classes', establishing that Things become members of Classes.

Outcome: **PASS**

CHARACTERISTIC: 56-28 with a Conceptual object

Reference: 56-28

Discourse sentence: "Consequently, Conceptual things are Conceptual objects with assigned Features."

Concept traceable: YES

Explanation: Sentence 56-28 states 'Conceptual things are Conceptual objects with assigned Features', establishing that Conceptual things have (are with) Conceptual objects.

Outcome: **PASS**

CHARACTERISTIC: 56-29 can conform to a Type

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-29 describes the semiotic triangle applied to concrete things discriminated by types; the discourse table records 'can conform to a Type' as the characteristic derived from this sentence. In context with surrounding sentences (Constraint 6), conformance to Types is the central mechanism illustrated.

Outcome: **PASS**

CHARACTERISTIC: 56-29 member of a conceived world

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-29 explicitly states things 'exist in a conceived world', establishing membership in a conceived world as a characteristic of intangible Concrete things.

Outcome: **PASS**

CHARACTERISTIC: 56-30 instance of a Type

Reference: 56-30

Discourse sentence: "The example referent in Fig. 5.3 (left) is a concrete thing of the type human."

Concept traceable: YES

Explanation: Sentence 56-30 states 'a concrete thing of the type human', directly naming the instance-of-Type relationship.

Outcome: **PASS**

CHARACTERISTIC: 56-32 collection of strings

Reference: 56-32

Discourse sentence: "The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course)."

Concept traceable: YES

Explanation: Sentence 56-32 refers to a 'name class that is used to refer to humans' in the context of the string 'Charlie Chaplin'; by concept primacy, a name class is a collection of strings.

Outcome: **PASS**

CHARACTERISTIC: 56-33 can be represented by a Conceptual thing

Reference: 56-33

Discourse sentence: "To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin')."

Concept traceable: YES

Explanation: Sentence 56-33 explicitly introduces the concrete thing <the human Charlie Chaplin> and the 'corresponding conceptual thing', establishing that a Concrete thing can be represented by a Conceptual thing.

Outcome: **PASS**

CHARACTERISTIC: 57-09 collection of numeric codes

Reference: 57-09

Discourse sentence: "The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things."

Concept traceable: YES

Explanation: Sentence 57-09 describes the name class that contains numeric codes (like '701'), establishing Name class as a collection of numeric codes.

Outcome: **PASS**

CHARACTERISTIC: 58-14 variable of Conceptual fact

Reference: 58-14

Discourse sentence: "The example conceptual fact is that the Conceptual thing signified by (object 701) is a human."

Concept traceable: YES

Explanation: Sentence 58-14 states the Conceptual thing signified by (object 701) is a variable of the conceptual fact 'is a human', directly establishing this characteristic.

Outcome: **PASS**

CHARACTERISTIC: 59-01 variable of Unary fact

Reference: 59-01

Discourse sentence: "If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity."

Concept traceable: YES

Explanation: Sentence 59-01 states 'If the thing of a Unary fact is a Concrete thing', directly establishing that Things can be variables of Unary facts.

Outcome: **PASS**

CHARACTERISTIC: 59-02 variable of Concrete unary fact

Reference: 59-02

Discourse sentence: "As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin"."

Concept traceable: YES

Explanation: Sentence 59-02 refers to 'the concrete unary fact that object 701 is a human', establishing Charlie Chaplin (object 701) as a variable of this Concrete unary fact.

Outcome: **PASS**

CHARACTERISTIC: 59-04 of time scale

Reference: 59-04

Discourse sentence: "As an example, day 2411108 is a value on the time scale Julian Date."

Concept traceable: YES

Explanation: Sentence 59-04 states 'day 2411108 is a value on the time scale Julian Date', establishing 'of time scale' as a characteristic of Abstract thing values.

Outcome: **PASS**

CHARACTERISTIC: 59-08 in reality

Reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Concept traceable: YES

Explanation: Sentence 59-08 explicitly states 'Concrete things are the things that we deal with in reality', establishing 'in reality' as a characteristic of Concrete things.

Outcome: **PASS**

CHARACTERISTIC: 59-09 tangible

Reference: 59-09

Discourse sentence: "As said, concrete things can be tangible (like a house) and intangible (like a rental)."

Concept traceable: YES

Explanation: Sentence 59-09 explicitly states 'concrete things can be tangible (like a house)', directly naming 'tangible' as a characteristic.

Outcome: **PASS**

CHARACTERISTIC: 59-09 intangible

Reference: 59-09

Discourse sentence: "As said, concrete things can be tangible (like a house) and intangible (like a rental)."

Concept traceable: YES

Explanation: Sentence 59-09 explicitly states concrete things can be 'intangible (like a rental)', directly naming 'intangible' as a characteristic.

Outcome: **PASS**

CHARACTERISTIC: 60-05 can be an instance of a Type

Reference: 60-05

Discourse sentence: "A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible)."

Concept traceable: YES

Explanation: Sentence 60-05 states 'A Type is defined as a template that the mind applies when we consider Concrete things', establishing that Concrete things can be instances of Types (are considered in relation to Types).

Outcome: **PASS**

CHARACTERISTIC: 60-06 with a Form

Reference: 60-06

Discourse sentence: "The Form of a Thing is the collection of its Features."

Concept traceable: YES

Explanation: Sentence 60-06 states 'The Form of a Thing is the collection of its Features', establishing that Things have (are with) a Form.

Outcome: **PASS**

CHARACTERISTIC: 60-08 may conform to a Type

Reference: 60-08

Discourse sentence: "In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform."

Concept traceable: YES

Explanation: Sentence 60-08 states the template is put over 'the things in the world, and sieves out the ones that conform', directly establishing 'may conform to a Type' for Things.

Outcome: **PASS**

CHARACTERISTIC: 60-08 member of a World

Reference: 60-08

Discourse sentence: "In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform."

Concept traceable: YES

Explanation: Sentence 60-08 refers to 'the things in the world', directly establishing that Things are members of a World.

Outcome: **PASS**

CHARACTERISTIC: 60-21 variable of an Unary fact

Reference: 60-21

Discourse sentence: "Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN."

Concept traceable: YES

Explanation: Sentence 60-21 states HUMAN is the Conceptual class of Things 'predicated by Facts of the Type human'; each such Thing is a variable of (= is predicated by) a Unary fact, establishing this characteristic.

Outcome: **PASS**

CHARACTERISTIC: 60-22 extension of a Type

Reference: 60-22

Discourse sentence: "This class is called the extension of the type human."

Concept traceable: YES

Explanation: Sentence 60-22 explicitly states 'This class is called the extension of the type human', directly naming 'extension of a Type' as the characteristic.

Outcome: **PASS**

CHARACTERISTIC: 60-23 with type intension

Reference: 60-23

Discourse sentence: "Conversely, the type human is called the intension of the class HUMAN."

Concept traceable: YES

Explanation: Sentence 60-23 explicitly states 'the type human is called the intension of the class HUMAN', establishing that the class HUMAN has a type as its intension.

Outcome: **PASS**

CHARACTERISTIC: 60-25 represents a Concrete class

Reference: 60-25

Discourse sentence: "Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN]."

Concept traceable: YES

Explanation: Sentence 60-25 explicitly states 'the conceptual class (HUMAN) represents the concrete class [HUMAN]', directly naming this characteristic.

Outcome: **PASS**

CHARACTERISTIC: 60-25 can be represented by a Conceptual class

Reference: 60-25

Discourse sentence: "Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN]."

Concept traceable: YES

Explanation: Sentence 60-25 explicitly states (HUMAN) represents [HUMAN], establishing by implication that [HUMAN] can be represented by a Conceptual class.

Outcome: **PASS**

CHARACTERISTIC: 61-04 variable of a Fact

Reference: 61-04

Discourse sentence: "A Fact starts to exist either as soon as the Form of a thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms."

Concept traceable: YES

Explanation: Sentence 61-04 states 'A Fact starts to exist either as soon as the Form of a thing conforms to a Type', establishing Things as variables of Facts.

Outcome: **PASS**

CHARACTERISTIC: 61-06 comes into existence

Reference: 61-06

Discourse sentence: "Through this event, a new entity of the type butterfly starts to exist."

Concept traceable: YES

Explanation: Sentence 61-06 states 'a new entity of the type butterfly starts to exist', directly naming the characteristic 'comes into existence'.

Outcome: **PASS**

CHARACTERISTIC: 61-06 part of an event

Reference: 61-06

Discourse sentence: "Through this event, a new entity of the type butterfly starts to exist."

Concept traceable: YES

Explanation: Sentence 61-06 states 'Through this event, a new entity... starts to exist', establishing that the coming into existence of a thing is part of an event.

Outcome: **PASS**

CHARACTERISTIC: 61-09 with Form

Reference: 61-09

Discourse sentence: "A Fact ends to exist either as soon as the Form of the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist."

Concept traceable: YES

Explanation: Sentence 61-09 refers to 'the Form of the Thing that the Fact is about', establishing that Things have (are with) a Form.

Outcome: **PASS**

CHARACTERISTIC: 61-12 goes out of existence

Reference: 61-12

Discourse sentence: "Consequently, the entity of the type caterpillar ends to exist."

Concept traceable: YES

Explanation: Sentence 61-12 states 'the entity of the type caterpillar ends to exist', directly naming the characteristic 'goes out of existence'.

Outcome: **PASS**

CHARACTERISTIC: 65-01 can be conceptualised

Reference: 65-01

Discourse sentence: "Figure 5.9 exhibits the conceptualisation of things of the type sit-on-able."

Concept traceable: YES

Explanation: Sentence 65-01 states 'the conceptualisation of things of the type sit-on-able', directly establishing that Things can be conceptualised.

Outcome: **PASS**

CHARACTERISTIC: 65-02 can have a role in a relationship

Reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Concept traceable: YES

Explanation: Sentence 65-02 states 'the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair)', establishing that Things can have roles in relationships.

Outcome: **PASS**

CHARACTERISTIC: 66-05 a collection of components

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: YES

Explanation: Sentence 66-05 describes Theseus' ship being preserved by replacing 'the old planks', thereby describing it as a composite (collection of components); the discourse entry records 'a collection of components' as the characteristic for Composite thing from 66-05.

Outcome: **PASS**

CHARACTERISTIC: 69-06 with properties

Reference: 69-06

Discourse sentence: "Instead they refer to different things, with different properties."

Concept traceable: YES

Explanation: Sentence 69-06 states things have 'different properties', directly establishing 'with properties' as a characteristic of Things.

Outcome: **PASS**

CHARACTERISTIC: 54-01 Thing

Reference: 54-01

Discourse sentence: "Thoughts are things in the mind."

Concept traceable: YES

Explanation: Sentence 54-01 introduces Thoughts as things in the mind, establishing the concept 'Thing' as a term from this foundational sentence.

Outcome: **PASS**

CHARACTERISTIC: 54-02 Concrete thing

Reference: 54-02

Discourse sentence: "They represent concrete things (i.e. things outside the mind), called referents."

Concept traceable: YES

Explanation: Sentence 54-02 introduces 'concrete things (i.e. things outside the mind)', directly establishing the term 'Concrete thing'.

Outcome: **PASS**

CHARACTERISTIC: 55-12 World

Reference: 55-12

Discourse sentence: "Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions."

Concept traceable: YES

Explanation: Sentence 55-12 introduces the concept of a World as defined by Bunge, establishing the term 'World'.

Outcome: **PASS**

CHARACTERISTIC: 56-20 Abstract thing

Reference: 56-20

Discourse sentence: "Abstract objects are Objects of Abstract things, like numbers and boolean values."

Concept traceable: YES

Explanation: Sentence 56-20 explicitly introduces the term 'Abstract things' with Abstract objects.

Outcome: **PASS**

CHARACTERISTIC: 56-21 Conceptual thing

Reference: 56-21

Discourse sentence: "When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing."

Concept traceable: YES

Explanation: Sentence 56-21 explicitly introduces the term 'Conceptual things' created by the mind.

Outcome: **PASS**

CHARACTERISTIC: 56-24 Class

Reference: 56-24

Discourse sentence: "It is called classifying: we observe similarities and differences between things and accordingly put them into different classes."

Concept traceable: YES

Explanation: Sentence 56-24 explicitly introduces the classification activity and the term 'classes', establishing the term 'Class'.

Outcome: **PASS**

CHARACTERISTIC: 56-29 Tangible concrete thing

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-29 introduces the subcategory of concrete things that are 'tangible', establishing the term 'Tangible concrete thing'.

Outcome: **PASS**

CHARACTERISTIC: 56-29 Intangible concrete thing

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-29 introduces the subcategory of concrete things that are 'intangible', establishing the term 'Intangible concrete thing'.

Outcome: **PASS**

CHARACTERISTIC: 56-29 Concieved world

Reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Concept traceable: YES

Explanation: Sentence 56-29 introduces the concept of 'a conceived world' as the domain of intangible things, establishing the term 'Conceived world'.

Outcome: **PASS**

CHARACTERISTIC: 56-32 Name class

Reference: 56-32

Discourse sentence: "The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course)."

Concept traceable: YES

Explanation: Sentence 56-32 explicitly introduces the term 'name class' as the class of strings used to refer to entities.

Outcome: **PASS**

CHARACTERISTIC: 57-02 Concrete class

Reference: 57-02

Discourse sentence: "The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans."

Concept traceable: YES

Explanation: Sentence 57-02 explicitly introduces the term 'Concrete class' via [HUMAN].

Outcome: **PASS**

CHARACTERISTIC: 60-21 Conceptual class

Reference: 60-21

Discourse sentence: "Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN."

Concept traceable: YES

Explanation: Sentence 60-21 explicitly introduces the term 'Conceptual class' as the class HUMAN of Conceptual things.

Outcome: **PASS**

CHARACTERISTIC: 66-05 Composite thing

Reference: 66-05

Discourse sentence: "The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place."

Concept traceable: YES

Explanation: Sentence 66-05 introduces the concept of a composite thing via Theseus' ship (a thing whose components are replaced). The discourse entry records 'Composite thing' as a term from this sentence.

Outcome: **PASS**

4. Check 3 Detail – Relationships with Explicit References

Objective: Verify that every marked cell in I with its own explicit key identifier can be traced to a discourse sentence supporting the item–characteristic relationship. Cells where the cell reference matches the item's own reference or the characteristic's own reference are -- # (inherited) and are not independently checked here.

RELATIONSHIP: 56-19 a human — 56-21 can be observed

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly states that tangible concrete things can be observed, and names humans as tangible ('humans and dogs' are given as examples of tangible things that 'one can observe'). This directly establishes 'a human | can be observed'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a human — 59-09 tangible

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly lists 'humans' as tangible concrete things ('tangible, so things that one can observe, like humans and dogs'), directly establishing 'a human | tangible'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a human — 54-01 Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: YES

Explanation: Sentence 59-08 states Concrete things are 'the things that we deal with in reality', and lists 'the people we cooperate with' as an example. By concept primacy, 'people' are humans; the sentence thereby establishes that humans (as people) are Things in reality. The discourse table records this key for the concept sign 'Thing' with example 'a human in reality', confirming the traceability.

Outcome: **PASS**

RELATIONSHIP: 56-19 a dog — 56-21 can be observed

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly names 'dogs' alongside humans as examples of tangible things that can be observed ('like humans and dogs'), directly establishing 'a dog | can be observed'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a dog — 59-09 tangible

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly names 'dogs' as tangible concrete things, directly establishing 'a dog | tangible'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a membership — 59-09 intangible

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly names 'memberships' as intangible concrete things that exist in a conceived world ('like memberships and rentals'), directly establishing 'a membership | intangible'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a rental — 59-09 intangible

Cell reference: 56-29

Discourse sentence: "Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals."

Relationship traceable: YES

Explanation: Sentence 56-29 explicitly names 'rentals' as intangible concrete things, directly establishing 'a rental | intangible'.

Outcome: **PASS**

RELATIONSHIP: 56-19 a rental — 54-01 Thing

Cell reference: 59-08

Discourse sentence: "Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer."

Relationship traceable: YES

Explanation: Sentence 59-08 mentions 'the rental contract of our house' as one of the Concrete things in reality. By concept primacy, 'a rental contract' is the same concept as 'a rental'. The discourse table records key 59-08 for concept sign 'Thing' with example 'a rental in reality'.

Outcome: **PASS**

RELATIONSHIP: 56-33 (the human Charlie Chaplin) — 56-30 instance of a Type

Cell reference: 58-14

Discourse sentence: "The example conceptual fact is that the Conceptual thing signified by (object 701) is a human."

Relationship traceable: YES

Explanation: Sentence 58-14 states 'the Conceptual thing signified by (object 701) is a human'. This directly establishes that the Conceptual thing (the human Charlie Chaplin) — i.e., the entity signified by (object 701) — is an instance of the type human, i.e., 'instance of a Type'.

Outcome: **PASS**

RELATIONSHIP: 59-02 Charlie Chaplin — 54-01 Thing

Cell reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory."

Relationship traceable: YES

Explanation: Sentence 65-02 refers explicitly to 'the Thing (Charlie Chaplin)', directly establishing that Charlie Chaplin is a Thing. This independent reference provides additional discourse support beyond the item's own introduction sentence.

Outcome: **PASS**

RELATIONSHIP: 59-08 a car — 66-05 a collection of components

Cell reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Relationship traceable: YES

Explanation: Sentence 66-26 states 'they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity', explicitly describing a car as having parts. The existence of named parts establishes a car as a composite with a collection of components, supporting 'a car | a collection of components'.

Outcome: **PASS**

RELATIONSHIP: 59-08 a car — 66-05 Composite thing

Cell reference: 66-26

Discourse sentence: "With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity."

Relationship traceable: YES

Explanation: Sentence 66-26 explicitly discusses the identity-bearing component of a car, placing cars in the same discourse context as composite things. The sentence establishes cars as having components (Composite things), directly grounding 'a car | Composite thing'. By concept primacy, the characterisation of cars as having identity-bearing parts is the defining property of a Composite thing.

Outcome: **PASS**

RELATIONSHIP: 59-08 a house — 56-29 Tangible concrete thing

Cell reference: 59-09

Discourse sentence: "As said, concrete things can be tangible (like a house) and intangible (like a rental)."

Relationship traceable: YES

Explanation: Sentence 59-09 explicitly states 'concrete things can be tangible (like a house)', naming a house as the canonical example of a tangible concrete thing, directly establishing 'a house | Tangible concrete thing'.

Outcome: **PASS**

RELATIONSHIP: 60-21 HUMAN — 56-24 Class

Cell reference: 60-23

Discourse sentence: "Conversely, the type human is called the intension of the class HUMAN."

Relationship traceable: YES

Explanation: Sentence 60-23 states 'the type human is called the intension of the class HUMAN', explicitly referring to HUMAN as a class. By concept primacy, this reference to 'the class HUMAN' directly establishes 'HUMAN | Class'.

Outcome: **PASS**

RELATIONSHIP: 66-05 a ship — 56-14 with Object

Cell reference: 66-13

Discourse sentence: "However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object."

Relationship traceable: YES

Explanation: Sentence 66-13 states 'Theseus' ship... has kept its Object', directly establishing that a ship has (is with) an Object.

Outcome: **PASS**

RELATIONSHIP: 66-05 a ship — 56-19 with Concrete object

Cell reference: 66-19

Discourse sentence: "We are now confronted with two concrete ships, each of them having its own Concrete object."

Relationship traceable: YES

Explanation: Sentence 66-19 states 'two concrete ships, each of them having its own Concrete object', directly establishing that a ship has a Concrete object.

Outcome: **PASS**

RELATIONSHIP: 66-05 a ship — 60-05 can be an instance of a Type

Cell reference: 66-10

Discourse sentence: "This argument is supported by the distinction between matter-constant (the ship is composed of the same part instances) and form-constant (the ship is an instance of the type ship), as proposed by Simons."

Relationship traceable: YES

Explanation: Sentence 66-10 explicitly states 'the ship is an instance of the type ship', directly establishing 'a ship | can be an instance of a Type'.

Outcome: **PASS**

RELATIONSHIP: 66-05 a ship — 60-08 may conform to a Type

Cell reference: 66-14

Discourse sentence: "This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before."

Relationship traceable: YES

Explanation: Sentence 66-14 states 'if the Thing would not conform to the Type ship anymore', directly establishing that a ship may conform to a Type (and by implication, may also cease to conform).

Outcome: **PASS**

RELATIONSHIP: 66-05 a ship — 54-02 Concrete thing

Cell reference: 66-19

Discourse sentence: "We are now confronted with two concrete ships, each of them having its own Concrete object."

Relationship traceable: YES

Explanation: Sentence 66-19 refers to 'two concrete ships', directly establishing that a ship is a Concrete thing by using the term 'concrete' in combination with 'ships'.

Outcome: **PASS**

RELATIONSHIP: 66-14 a butterfly — 56-30 instance of a Type

Cell reference: 61-06

Discourse sentence: "Through this event, a new entity of the type butterfly starts to exist."

Relationship traceable: YES

Explanation: Sentence 61-06 states 'a new entity of the type butterfly starts to exist', explicitly naming the type butterfly and establishing that a butterfly is an instance of this type, directly grounding 'a butterfly | instance of a Type'.

Outcome: **PASS**

RELATIONSHIP: 66-14 a caterpillar — 56-30 instance of a Type

Cell reference: 61-12

Discourse sentence: "Consequently, the entity of the type caterpillar ends to exist."

Relationship traceable: YES

Explanation: Sentence 61-12 states 'the entity of the type caterpillar ends to exist', explicitly naming the type caterpillar and establishing that a caterpillar is an instance of this type.

Outcome: **PASS**

5. Check 4 Detail – Completeness of the Formal Context

Objective: Identify items, characteristics, or relationships present in the discourse but absent from the formal context. Type application and constraint-based derivation are strictly forbidden; only explicit discourse sentences are admissible.

MISSING RELATIONSHIP:

FOUND: (the class HUMAN) — Conceptual thing

Discourse reference: 57-03

Discourse sentence: "It is represented in our mind by the thought (the class HUMAN)."

Explanation: Sentence 57-03 states '(the class HUMAN)' is represented 'in our mind by the thought'. By concept primacy (Constraint 1, retained in Check 4), 'a thought in our mind' is the defining expression of a Conceptual thing (cf. 54-01: 'Thoughts are things in the mind'; 59-10: 'Conceptual things are things in the mind'). The sentence therefore explicitly places (the class HUMAN) within the concept of Conceptual thing without requiring type application or constraint-based derivation — the reference is direct through synonymous phrasing. The formal context should mark the cell (the class HUMAN) × Conceptual thing.

Outcome: **FOUND**

MISSING ITEM:

FOUND: business entity (or: business thing)

Discourse reference: 68-09

Discourse sentence: "The same 'business entity' (upper part) appears to conform to two types, namely to the type purchase (left side), and to the type sale (right side), respectively applied by the buyer and the seller enterprise."

Explanation: Sentence 68-09 explicitly introduces 'the same business entity' as a concrete example of a Thing that conforms to two types simultaneously (purchase and sale). This is a distinct example not present in the formal context: no item in G represents a business entity or business thing. The sentence explicitly names the example and attributes to it the characteristic 'may conform to a Type' (which is in M under ref 60-08), and by extension 'can be an instance of a Type'. Given that the formal context includes items such as 'a student', 'a butterfly', and 'a caterpillar' for their conceptual significance, the business entity — which uniquely illustrates dual-type conformance — should be included as an item in G. No other item in G exemplifies this property, making the omission noteworthy.

Outcome: **FOUND**

6. Overall Verdict

Items (Check 1): 35 checked | 35 PASS | 0 FAIL | 0 SKIPPED

Characteristics (Check 2): 59 checked | 59 PASS | 0 FAIL | 0 SKIPPED

Relationships (Check 3): 21 independently checked | 21 PASS | 0 FAIL | 0 SKIPPED

Completeness (Check 4): 2 addition(s) identified in the discourse.

OVERALL TRACEABILITY ASSESSMENT: FULLY TRACEABLE

All 35 items, all 59 characteristics, and all 21 independently-referenced relationships pass their respective traceability checks. Every entry in the formal context can be grounded in a specific discourse sentence from Enterprise Ontology (Dietz & Mulder, 2024), in accordance with the Core Validation Principle (concept primacy over symbol) and Constraints 1–7.

COMPLETENESS: PARTIALLY COMPLETE

Check 4 identified 2 additions that should be present in the formal context:
1. MISSING RELATIONSHIP: (the class HUMAN) × Conceptual thing — Sentence 57-03 explicitly places (the class HUMAN) as 'a thought in our mind', which by concept primacy equates to Conceptual thing. This relationship should be marked in the formal context.
2. MISSING ITEM: business entity (business thing) — Sentence 68-09 explicitly introduces a business entity as an example of a Thing conforming to two types simultaneously. This distinctive example is absent from G and should be added, with the characteristic 'may conform to a Type' marked.